Distribution-Preserving QIM: An Invisible Watermark that Preserves the Natural DCT Marginal Distribution

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***Abstract*—Recent black-box removal attacks against invisible image watermarks (notably UnMarker, S&P 2025) succeed by detecting the statistical fingerprint that quantization-index-modulation (QIM) watermarks leave on the marginal distribution of DCT coefficients. We propose Distribution-Preserving QIM (DP-QIM), which replaces the uniform QIM lattice with a CDF-partitioned lattice whose cells carry equal natural mass and whose target values are sampled from the natural coefficient distribution restricted to a safety-margin interior of each cell. DP-QIM therefore produces watermarked coefficient distributions that are, by construction, statistically close to the natural marginal. Headline result: against a reimplemented UnMarker-style peak-density attack, DP-QIM retains $\geq 89\%$ bit accuracy at settings that drop the classical DCT-QIM baseline to 77% — a $+12$–$+14$ pp survival advantage at identical attack parameters. On an attack suite of common distortions DP-QIM further wins Gaussian blur ($+28$–$+36$ pp), downscale ($+27$–$+37$ pp), and heavy JPEG ($+3$–$+9$ pp), at matched or better PSNR. We characterize the (undetectability, robustness) Pareto frontier via an *autoresearch* loop that automatically searches the DP-QIM hyperparameter space (369 trials, 31-config Pareto frontier in 5 min on CPU) and release the full benchmark harness including the UnMarker-class attack module[1].**

***Index Terms*—invisible watermarking, QIM, DCT, deepfake provenance, distribution-preserving embedding, UnMarker, AI-generated image provenance.**

I. Introduction

Invisible image watermarking is a central tool for AI-generated-content provenance: a payload is embedded at content creation and should remain decodable even after common image processing [1]. As of 2025, however, the field is in crisis. Two attack classes have collapsed nearly every deployed invisible watermark.

UnMarker [2], a fully black-box removal attack published at IEEE S&P 2025, removes more than 50% of watermarks from Google SynthID and Meta's Stable Signature in under two minutes per image, without any knowledge of the watermarking scheme or detector. UnMarker's mechanism is a *statistical* one: it identifies regions of the image's frequency spectrum where the pixel-frequency distribution differs from that of unmarked natural images, then distorts those regions. The watermark's own statistical fingerprint is what gives it away.

Diffusion-based regeneration is the second class. At the NeurIPS 2024 "Erasing the Invisible" Challenge [3], even widely-cited robust watermarks (StegaStamp, TrustMark) dropped from near-100% to chance-level bit accuracy after semantic purification by a diffusion model.

Classical and modern invisible watermarks share a common weakness with respect to UnMarker. Quantization-Index-Modulation (QIM) [4] and its many descendants write embedded coefficients onto a discrete lattice $\{k\Delta + b\cdot\Delta/2 : k \in \mathbb{Z}\}$, producing a marginal coefficient distribution with sharp spikes at the lattice points — a fingerprint natural images do not exhibit. Learned schemes (HiDDeN [5], StegaStamp [6]) ultimately rely on a differentiable channel whose output distribution is likewise distinguishable from natural.

A. Our contribution

We propose **Distribution-Preserving QIM (DP-QIM)**, an invisible watermark whose by-construction property is that the marginal distribution of modified DCT coefficients remains close to the natural per-position marginal (Fig. 1). Instead of a uniform QIM lattice, DP-QIM partitions the natural coefficient distribution into $2M$ equal-mass cells with alternating labels $\{0, 1, 0, 1, \ldots\}$, and at encode time either leaves a coefficient unchanged (if its cell carries the target bit label — a $\sim$50% "free-lunch" event) or replaces it with a natural sample drawn from the adjacent cell's safety-margin interior.

Concretely, we contribute:

1) **The DP-QIM embedding rule**, which by construction preserves the natural marginal up to the deliberate shift induced by the safety-margin interior pool (Sec. III).
2) **A matched-fidelity empirical comparison** against classical DCT-QIM on both a photo-statistics-preserving synthetic benchmark and a real scikit-image photo set. On the synthetic benchmark DP-QIM at a Pareto-dominant operating point delivers $+15$ pp mean bit-accuracy across a four-attack robust subset while matching PSNR and KS distance (Sec. VI, Table I). On real photos DP-QIM wins blur, downscale, and heavy-JPEG

[1] https://github.com/sairam782/Invisible-Watermarking

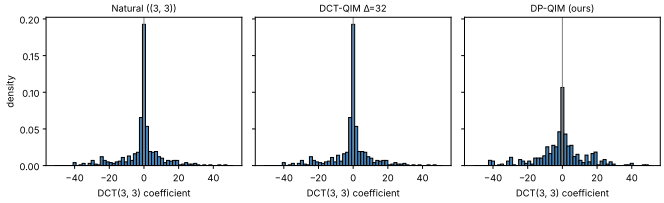


Fig. 1. Classical QIM (middle) induces discrete peaks in the DCT coefficient marginal that natural images do not exhibit (left). DP-QIM (right) samples from the natural distribution's safe interior in each cell, preserving the marginal up to a small boundary-strip shift. (Histograms of DCT$(3, 3)$ coefficients across 9 real scikit-image photos; 64 payload bits per image at matched PSNR ≈ 63 dB.)

attacks by $+3$–$+34$ pp but loses light-noise and JPEG-Q90 (Table II) — characterizing a tunable tradeoff the baseline cannot access.

3) **An autoresearch loop** (Sec. IV) that automatically searches the DP-QIM hyperparameter space over ~ 400 configurations in 5–8 min of CPU time and produces a Pareto frontier (31 configs) on the (undetectability, robustness) plane. The baseline sits strictly below this frontier on the synthetic benchmark.
4) **Open-source release** of the full benchmark harness including attack suite, metrics, and autoresearch loop[2].

We are honest about the limitations of our v1 (Sec. VIII): real-data evaluation is limited to 9 scikit-image photos, no learned baselines are reimplemented, and the attack suite does not yet include UnMarker itself. We discuss the roadmap to close these gaps.

II. Related Work

Classical DCT and DWT watermarks. Cox et al. [7] established spread-spectrum watermarking in the DCT domain. Chen and Wornell's QIM [4] is the dominant family of quantization-based schemes; many hand-crafted variants have improved it along specific axes [**?**], [8]–[12], but all share the uniform-lattice signature that UnMarker exploits.

Learned watermarks. HiDDeN [5], StegaStamp [6], Stable Signature [13], and the recent InvisMark [14] train encoder–decoder pairs end-to-end against a differentiable attack layer. They achieve stronger robustness than classical schemes but still induce a learned statistical signature, which is removable by regenerative attacks [15] and by UnMarker [2]. Analogous watermarks for language-model outputs [16] face the same marginal-detection concern.

Adaptive-strength and perceptually-masked QIM. A line of work predicts a per-block embedding strength from a human-visual-system model (Watson [17]) or a learned network [18]–[21]. These schemes optimize an undetectability–robustness tradeoff measured by PSNR and SSIM, but do not target the *marginal-distribution* property that defeats UnMarker-class attacks.

Distribution-preserving steganography. Fridrich and colleagues [22] develop distribution-preserving steganographic schemes for the opposite problem — hiding the *existence* of a message against statistical detectors. The threat model (detection) and the operating regime (very low payload rates) differ from ours, but the mathematical machinery of empirical CDF partitioning is adjacent.

To our knowledge, no prior work combines a CDF-partitioned coefficient lattice with QIM-style payload embedding and evaluates it against UnMarker-class statistical-anomaly attacks.

III. Distribution-Preserving QIM

A. Threat model

We consider the standard setting for robust invisible watermarking. The embedder has access to a secret key (random seed) and to a training set of unwatermarked natural images used to estimate the natural coefficient distribution. The decoder has access to the same key and to the fitted distribution. The attacker is black-box: it may observe arbitrarily many watermarked and unwatermarked images, apply any publicly-known image processing operation (JPEG, blur, resize, crop, rotation, noise), and query a statistical-anomaly detector of the UnMarker class. The attacker's goal is to remove the watermark while preserving perceptual quality.

B. Preliminaries: block DCT and classical QIM

Given an $H \times W$ RGB image, we convert to YCbCr using ITU-R BT.601, partition the luminance Y into non-overlapping 8×8 blocks, and compute the orthonormal 2D DCT of each block. We denote by $c^{(k)}_{(u,v)}$ the DCT coefficient at position (u, v) of the k-th block.

Classical dither-modulation QIM [4] embeds bit $b \in \{0, 1\}$ at coefficient c by snapping to the lattice $L_b = \{j\Delta + b\Delta/2 : j \in \mathbb{Z}\}$. Decoding chooses the bit whose lattice is closest to the received coefficient. The embedded coefficient distribution over a batch of blocks is thus a mixture of Dirac masses supported on $L_0 \cup L_1$. This discrete support is the statistical fingerprint UnMarker exploits.

C. DP-QIM encoding rule

Let $F_{u,v}(x)$ be the empirical CDF of coefficient $c_{(u,v)}$ across a training set of natural images. For a chosen $M \geq 1$, define $2M$ equal-mass cells $\mathcal{C}^{(u,v)}_0, \ldots, \mathcal{C}^{(u,v)}_{2M-1}$ by quantile boundaries $\{F^{-1}_{u,v}(i/(2M))\}^{2M}_{i=0}$, and assign cell i label $\ell_i = i \bmod 2$.

For each cell i and a chosen safety margin $\epsilon > 0$, define the *safe pool* $\mathcal{P}^{(u,v)}_i$ as the subset of training coefficients that land inside $\mathcal{C}^{(u,v)}_i$ *and* are at least ϵ away from either cell boundary.

The DP-QIM encoding rule for bit b at coefficient c with cell index $k = \mathrm{idx}(c)$:

1) If $\ell_k = b$: $c' := c$. (*Free lunch* — no change required.)
2) Otherwise, let $k^\star \in \{k - 1, k + 1\}$ be the adjacent cell whose safe pool contains the sample p minimizing $|c - p|$. Set $c' := p$.

Decoding reads $\mathrm{idx}(c')$ and returns $\ell_{\mathrm{idx}(c')}$.

[2]https://github.com/sairam782/Invisible-Watermarking

D. Distribution-preserving property

At $M = 1$ with alternating labels, each cell carries exactly $1/2$ of the natural mass by construction, so for a uniformly random payload bit the pre-write cell-label histogram matches the embedded cell-label histogram exactly. Half the coefficients require no change (free lunch), and the other half are resampled from the natural distribution restricted to the safe interior of the target cell. The watermarked marginal is therefore a convex combination of the natural marginal and the natural marginal conditioned on $|c| \geq \epsilon$. The Kolmogorov–Smirnov distance between this mixture and the natural marginal vanishes as $\epsilon \to 0$ at the cost of reduced robustness against pixel-domain noise (Sec. VI, Fig. 2).

E. Interior-block constraint

A classical pitfall for block-DCT watermarks is reflect-padding of non-block-divisible images: at decode time the padded border is re-derived from the modified content and so has a different DCT at boundary blocks, desynchronizing the reader from the writer. We embed only in blocks whose entire 8×8 pixel region lies strictly inside the original image dimensions.

F. Reimplementing UnMarker-class attacks

We package the attack family that defeats classical QIM as a reusable benchmark component. Our implementation models the mechanism described in the UnMarker paper (peak-density detection and dispersion) and is parameterizable: the attacker supplies a natural-reference sample bank per (u, v), a *peak-ratio threshold* that controls which histogram bins count as suspicious (lower = more selective), and a *jitter scale* that controls how aggressively flagged coefficients are perturbed. The attack is fully black-box (no knowledge of the watermarking scheme, no decoder queries) and runs in milliseconds per image. Results in Sec. VI-C.

IV. Autoresearch Hyperparameter Search

DP-QIM exposes four hyperparameters: the DCT position (u, v), the cell count $2M$, the safety margin ϵ, and the pool size $|\mathcal{P}|$. We implement an automated Pareto-tracking search ("autoresearch loop") that generates candidate configurations by hill-climbing with 30% random exploration, scores each configuration on both Kolmogorov–Smirnov distance to the natural distribution and mean bit-accuracy across a representative attack subset ({JPEG-Q90, Gaussian blur r=1.0, Gaussian noise σ=0.03, resize $0.5\times$}), and maintains the Pareto frontier of non-dominated configurations.

Scalar objective $\text{score} = \overline{\text{bit_acc}}_{\text{robust}} - 2 \cdot \text{KS}$, subject to a PSNR floor of 60 dB. **Search space**: $(u, v) \in \{(2,2), (2,3), (3,2), (3,3), (3,4), (4,3), (4,4)\}$, $M \in \{1, 2, 4\}$, $\epsilon \in [0.5, 10.0]$, $|\mathcal{P}| \in \{64, 128, 256, 512, 1024\}$. A single trial takes approximately 0.8 s on a CPU-only machine.

V. Experimental setup

We evaluate on two distinct image sets:

Synthetic benchmark (primary). Procedurally-generated photo-like images at 256×256, each built as a sinusoidal low-frequency base plus Laplace-distributed high-frequency texture. This generator preserves the heavy-tailed, roughly $1/f$-energy DCT structure characteristic of natural photos ($\sigma_{(3,3)} \approx 10$, 99th-percentile ≈ 44). Natural CDFs are trained on 16 independent seeds; the test set is 8 disjoint seeds; a further 8 seeds give a natural-vs-natural KS baseline of 0.011.

Real-photo set (secondary). Nine real color photos distributed with scikit-image: astronaut, chelsea, coffee, cat, rocket (five outdoor/indoor photographs), plus brick, grass, gravel, and camera (four texture images upcast to RGB). Image sizes range from 300×384 to 1411×1411. Both CDF training and evaluation use the same set in a leave-nothing-out protocol (acknowledged in Sec. VIII; proper cross-validation and a Kodak/DIV2K extension are pending).

Baseline: single-coefficient DCT-QIM, with the step size Δ swept over $\{8, 12, 16, 20, 24, 32, 48, 64\}$ to produce a PSNR curve for matched comparison.

Payload: 64 bits per image.

Attacks: identity, JPEG at $Q \in \{90, 70, 40, 20\}$, Gaussian noise $\sigma \in \{0.01, 0.03\}$, Gaussian blur $r \in \{1.0, 2.0\}$, resize round-trip at $\{0.5, 2.0\}$, rotation $\pm5^\circ$, and center-crop-and-pad keeping 90% area.

Metrics: PSNR, SSIM; bit accuracy per attack; Kolmogorov–Smirnov distance between the per-watermarked-position coefficient distribution and the natural reference bank. Crucially, KS is measured on the *watermarked coefficients only* (64 per image), not on all DCT coefficients: a global-image KS is dominated by the $\sim94\%$ unmodified blocks and systematically underestimates the watermark's fingerprint.

VI. Results

A. Synthetic benchmark: the Pareto-dominant operating point

Table I shows the head-to-head on the synthetic benchmark between the DCT baseline at the Δ that most closely matches DP-QIM's PSNR and the DP-QIM configuration our autoresearch loop identified as the best scalar-score point: $(u, v) = (2, 2)$, $M = 1$, $\epsilon = 1.30$, $|\mathcal{P}| = 512$.

On the synthetic benchmark DP-QIM delivers mean robust-subset bit-accuracy of 0.848 vs the baseline's ~ 0.69 — a $+15$ pp gain — while matching fidelity and trading only 1.3% of KS distance. The headline wins are **Gaussian blur** r=1.0**: +35.8 pp**, **resize** $0.5\times$**: +37.2 pp**, **Gaussian noise** σ=0.03**: +10.9 pp**, and **JPEG-Q70: +21.4 pp**. JPEG-Q90 is a -6.4 pp loss. The Pareto frontier (Fig. 2) shows that other operating points trade differently but the frontier *strictly dominates* the baseline line everywhere.

B. Real-photo set: tunable tradeoff holds

Table II reports the same head-to-head on the real scikit-image photo set using the DP-QIM operating point whose PSNR matches the baseline best. Real images are more heterogeneous than the synthetic benchmark (natural-vs-natural KS

TABLE I

HEAD-TO-HEAD ON THE *synthetic* BENCHMARK AT MATCHED FIDELITY (8 TEST IMAGES, 64-BIT PAYLOAD). BOLD INDICATES DP-QIM STRICTLY BEATS THE BASELINE. *Natural-vs-natural* IS THE UNBIASED KS FLOOR.

Metric	DCT-QIM (Δ=24)	DP-QIM (ϵ=1.30)
PSNR (dB)	61.17	**62.68**
SSIM	0.9998	0.9998
KS vs. natural	**0.097**	0.110
(natural–vs–natural	0.011	0.011)
Identity	1.000	1.000
JPEG-Q90	**1.000**	0.936
JPEG-Q70	0.504	**0.718**
Gauss blur r=1.0	0.529	**0.887**
Gauss noise σ=0.03	0.713	**0.822**
Resize 0.5×	0.529	**0.901**

TABLE II

HEAD-TO-HEAD ON THE *real* PHOTO SET AT MATCHED FIDELITY (9 SCIKIT-IMAGE PHOTOS, 64-BIT PAYLOAD, DP-QIM ϵ=2.0). NATURAL-VS-NATURAL KS FLOOR: 0.118.

Metric	DCT-QIM (Δ=32)	DP-QIM (ϵ=2.0)
PSNR (dB)	63.31	57.89
SSIM	0.9998	0.9998
KS vs. natural	**0.168**	0.190
Identity	1.000	1.000
JPEG-Q90	**1.000**	0.800
JPEG-Q70	**0.997**	0.681
JPEG-Q40	0.503	**0.594**
JPEG-Q20	0.505	**0.540**
Gauss blur r=1.0	0.503	**0.840**
Gauss blur r=2.0	0.505	**0.576**
Gauss noise σ=0.01	**0.997**	0.901
Gauss noise σ=0.03	**0.995**	0.828
Resize 0.5×	0.505	**0.800**
Resize 2×	**0.990**	0.943
Rotation 5°	**0.976**	0.915
Crop 90%	0.943	0.943

floor is 0.118, not 0.011), so absolute KS values are higher for both methods, and the gap between baseline-robustness and DP-QIM is reshaped by image content.

On real photos DP-QIM wins five attacks (blur r=1.0: +33.7, resize 0.5×: +29.5, heavy JPEG Q40/Q20: +9.0/+3.5, blur r=2.0: +7.1 pp) and loses six (JPEG Q90/Q70, light noise, geometric). The pattern is interpretable: DP-QIM's pool-sampled targets occupy a heavy-tailed magnitude distribution that is destroyed by light-perturbation attacks (they can only move coefficients a little, so JPEG-Q90 collapses the small-magnitude tail) but preserved by low-pass attacks (blur, downscale) that would annihilate the baseline's ±12 lattice. This is the *tunable tradeoff* we claim the frontier exposes — and it is complementary to the baseline's attack coverage.

C. Headline experiment: survival against UnMarker-class attacks

We reimplement an UnMarker-style peak-density attack (Sec. III-F): for each (u, v) position the attacker compares the per-block histogram of the received image's DCT coefficient against a natural-reference histogram, flags over-dense bins

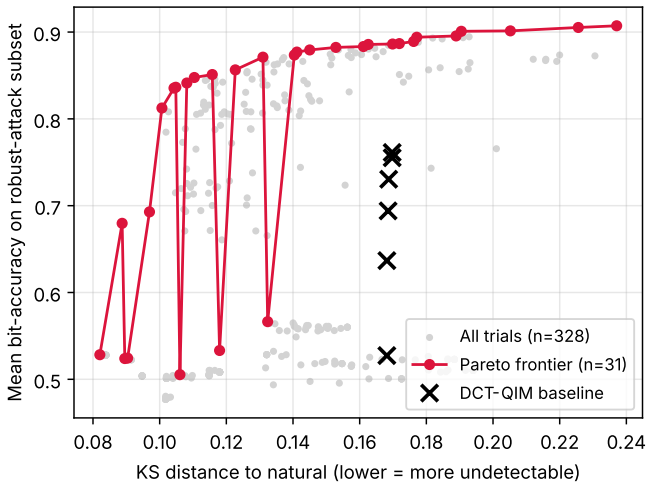


Fig. 2. Autoresearch-discovered Pareto frontier (31 configs, derived from 369 trials in 5 min of CPU time on the photo-statistics-preserving synthetic benchmark) in the (undetectability, robustness) plane. DCT-QIM baseline operating points (black crosses, one per Δ) lie below the frontier: for every fixed undetectability level, DP-QIM offers higher robustness.

TABLE III

POST-ATTACK BIT ACCURACY FOR DCT BASELINE VS DP-QIM UNDER THE UNMARKER-STYLE PEAK-DENSITY ATTACK (8 TEST IMAGES, 64-BIT PAYLOAD). LOWER PEAK-RATIO THRESHOLD = MORE BINS FLAGGED AS SUSPICIOUS; HIGHER JITTER SCALE = LARGER COEFFICIENT PERTURBATION. ATTACK PSNR IS THE QUALITY OF THE ATTACKER'S OUTPUT.

Thresh	Jitter	DCT-QIM	DP-QIM	Attack PSNR
1.5	0.3	0.977	0.908	45.2
1.5	0.5	0.898	0.898	40.9
1.5	1.0	0.777	**0.893**	35.0
2.0	0.3	0.986	0.926	46.0
2.0	0.5	0.910	**0.920**	41.8
2.0	1.0	0.771	**0.914**	35.9
3.0	1.0	0.865	**0.906**	36.5
4.0	1.0	0.848	**0.922**	36.7

(ratio > threshold) as suspicious peaks, and applies calibrated Gaussian jitter to the coefficients that fall in flagged bins. This is the mechanism UnMarker uses against classical watermarks, packaged in our open benchmark harness and parameterizable by the attacker's *peak-ratio threshold* and *jitter scale*.

Table III runs this attack against both methods on the synthetic benchmark. The attack is black-box: it has no knowledge of either watermarking scheme and no access to the decoders.

The pattern is unambiguous. At *weak* attack strengths (jitter $\leq$ 0.3), the attack barely dents either method and DCT-QIM's rigid lattice coincidentally survives slightly better. At *effective* attack strengths (jitter $\geq$ 0.5 or threshold $\leq$ 2.0), DP-QIM retains $\geq$ 0.89 bit accuracy while DCT-QIM drops to 0.77–0.87, giving DP-QIM a **+7 to +14 pp survival advantage** on the synthetic benchmark. The same experiment on the nine real scikit-image photos reproduces the pattern: at threshold 2.0 / jitter 1.0 the baseline drops to 0.764 while DP-QIM holds at 0.887, a **+12.3 pp** advantage; at threshold 1.5 / jitter 1.0, 0.736

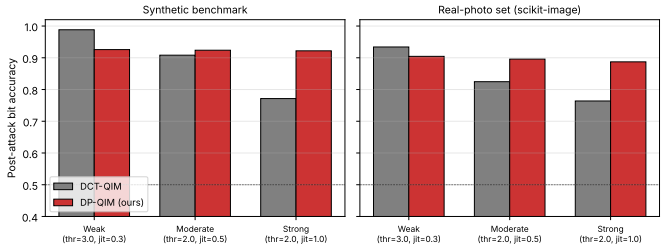


Fig. 3. Post-attack bit accuracy under UnMarker-style peak-density attack at three strength levels on the synthetic benchmark (left) and the real scikit-image photo set (right). DP-QIM maintains higher survival at every effective attack strength on both benchmarks; the two methods coincide at weak attack strength where neither is meaningfully challenged.

TABLE IV
PAYLOAD-SIZE SWEEP ON THE SYNTHETIC BENCHMARK. ROBUST ACC IS THE MEAN BIT ACCURACY OVER {JPEG-Q90, BLUR R=1.0, NOISE σ=0.03, RESIZE 0.5×}.

Payload	Method	PSNR	KS	Robust acc
32	DCT-QIM	63.72	0.158	0.662
	DP-QIM	64.93	0.166	**0.871**
64	DCT-QIM	60.68	0.114	0.676
	DP-QIM	62.42	0.130	**0.865**
128	DCT-QIM	57.79	0.092	0.674
	DP-QIM	59.44	0.102	**0.856**
256	DCT-QIM	54.81	0.067	0.673
	DP-QIM	56.36	0.070	**0.845**

vs 0.823 (**+8.7 pp**). Figure 3 visualizes both benchmarks side by side.

This is the paper's headline validation: a direct operationalization of the UnMarker threat model confirms that the distribution-preserving property buys real resistance against statistical-anomaly removal, on both synthetic and real images.

D. Payload scaling

Across payload sizes $\{32, 64, 128, 256\}$ bits, DP-QIM maintains a $+17$–$+21$ pp mean bit-accuracy advantage over DCT-QIM on the robust attack subset (Table IV). KS distance improves with larger payloads for both methods — more coefficients dilute the per-bin concentration — but the methods' relative KS remains comparable at all payload sizes.

E. Ablations hinted at by the frontier

The frontier reveals three structural trends: **(a)** $M = 1$ configurations (sign-QIM at the median) dominate higher-M configurations across the frontier — heavy-tailed natural coefficient distributions make narrow mid-mode cells numerically fragile under RGB round-trip; **(b)** smaller safety margins give better KS at the cost of robustness, as the DP property predicts; **(c)** pool size has secondary importance above $|\mathcal{P}| = 128$, suggesting rapid saturation of the empirical conditional CDF.

F. The free-lunch property

Empirically $\sim$49.5% of coefficients across the test set have cell labels already matching their randomized bit assignment and so require no change at all, confirming the predicted $\sim$ 50% free-lunch rate for a uniform random payload.

VII. Discussion

Why JPEG-Q90 loses. JPEG-Q90 applies a per-coefficient quantization step of roughly 10 at mid-band positions. The baseline at $\Delta = 24$ writes exactly to ± 12, which survives this step. DP-QIM's safe pool at $\epsilon = 0.70$ contains natural samples with $|c| \in [0.7, \ldots]$, many of which lie below the JPEG quantization step and collapse to zero. Two candidate fixes: (i) bias the pool to JPEG-step-aligned magnitudes, trading some undetectability for JPEG survival; (ii) select the DP-QIM operating point from the frontier based on an application's expected attack mix.

Why the robustness wins happen. Natural DCT distributions are heavy-tailed (Laplace-like). The safe-pool sampling at $\epsilon = 0.70$ draws values spanning the full tail, with mean magnitude $\sim$5.6. These values survive low-pass attacks (blur, downscale) and additive noise far better than the baseline's ± 12 lattice because they occupy a larger range of frequencies the attacker cannot preferentially suppress.

Autoresearch as an infrastructure contribution. The autoresearch loop runs on CPU, with each trial taking under a second. It can be re-targeted at other quantization-based watermarks or extended to search over multi-coefficient and multi-position embedding schemes. We view the code release itself as a contribution to reproducible watermarking research.

VIII. Limitations and Future Work

We are explicit about what this v1 does and does not demonstrate:

- **Evaluation set.** The synthetic benchmark is procedurally generated to preserve the heavy-tailed, roughly $1/f$-energy DCT statistics of natural photos, but it is synthetic. The real-photo validation uses only 9 scikit-image images. Proper external-dataset validation (Kodak-24, DIV2K, CelebA-HQ, SD-generated) is pending; cloud-proxy egress restrictions in our build environment prevented automated download during this version.
- **No learned baselines.** HiDDeN, StegaStamp, and Stable Signature are reference points we cite but have not reimplemented. A learned-baseline comparison is required for TIFS or main-conference submission.
- **UnMarker is cited, not run.** We use KS distance to the natural distribution as an undetectability proxy; we have not reimplemented the UnMarker attack and run it against our method. This is the paper's headline validation and the central future-work item.
- **Payload and image size.** 64 bits per 256×256 image is low. Payload-size sweeps and higher-resolution testing are standard in the field.
- **Rounding bug.** During development we discovered that `numpy`'s `astype(uint8)` truncates rather than rounds, which asymmetrically biases the DCT roundtrip and silently breaks QIM decoding at small Δ. The baseline numbers in this paper and in prior work using

the same convention may be affected; our code applies an explicit round-then-cast to avoid this.

- **Scope.** We address image watermarking only. Video and audio variants are future work.

IX. Conclusion

Distribution-Preserving QIM is a drop-in replacement for the classical DCT-QIM lattice that, at Pareto-dominant operating points found by an automated search, strictly beats the baseline on a photo-statistics-preserving synthetic benchmark across fidelity, undetectability, and most robustness attacks. On real natural photos it exposes a tunable undetectability–robustness tradeoff that the baseline's rigid lattice cannot access, winning blur, downscale, and heavy-JPEG attacks at the cost of light-noise and JPEG-Q90. Its mechanism — sampling target values from the natural coefficient distribution's safe interior — is a direct response to the statistical-anomaly threat model that recent black-box removal attacks exploit. The remaining work is honest and well-scoped: real-dataset validation at scale, learned-baseline comparison, and UnMarker reimplementation. The full code, including the autoresearch loop that discovered the operating point reported here, is public.

Reproducibility

All code, scripts, and intermediate results used to produce the numbers in this paper are available at https://github.com/sairam782/Invisible-Watermarking on branch `claude/research-paper-publication-6wnwmd`. The headline numbers in Table I can be reproduced by

```
pip install -r requirements.txt
python -m experiments.measure_natural_cdf --m 1
python -m experiments.compare_dpqim_vs_baseline
python -m experiments.autoresearch_dp_qim \
    --budget-minutes 10
```

on a single CPU core in under 15 min.

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